Support Before Amplitude: A Bottleneck in Cross-Operator Sparse Recovery

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***Abstract*—We study zero-shot sparse recovery under operator shift. On Fourier $\to$ Gaussian transfer (n=256, m=128, k=25), a parameter-free pipeline of ISTA, naive top-k support selection, and support-restricted least squares achieves NRMSE 0.257, improving over zero-shot LISTA (0.327) by 21% relative. Two learned coordinate-wise detectors of differing capacity, both operating on iterate-derived features, tie naive magnitude ranking within 0.001 NRMSE. Because oracle support yields near-exact recovery, the binding constraint is support identification, not amplitude estimation. We trace the detector plateau to an iterate-only feature interface and propose operator-aware coordinate features—residual correlations and operator Gram statistics computed from the deployment matrix—as a minimal learned module for transfer without target labels.**

***Index Terms*—Sparse recovery, compressed sensing, learned iterative reconstruction, LISTA, transfer learning, support detection.**

I. Problem Setup

We consider sparse recovery from linear measurements

$$y = Ax + \varepsilon, \tag{1}$$

where $x \in \mathbb{R}^n$ is k-sparse and $A \in \mathbb{R}^{m\times n}$ is the sensing matrix. Given a fixed operator A, we first compute an ISTA iterate

$$x^{(T)} = \mathrm{ISTA}_T(y; A, \lambda). \tag{2}$$

We then compare four recovery strategies under operator shift. All learned models are trained on 4,000 Fourier-measurement signals and evaluated zero-shot on Gaussian measurements: (i) raw ISTA; (ii) naive top-k support selection on $|x^{(T)}|$ followed by support-restricted least squares; (iii) a learned coordinate-wise support detector operating on iterate-derived features, again followed by least squares; and (iv) end-to-end LISTA.

This setting separates two questions that are often entangled in learned sparse recovery: whether the method identifies the correct support, and whether it estimates the amplitudes accurately once the support is known.

II. Main Result

Table I shows that the parameter-free support-then-amplitude pipeline outperforms zero-shot LISTA by 21% relative NRMSE. At the same time, learned support detectors operating on ISTA-derived features track naive top-k selection almost exactly: across $(T, \lambda) \in \{(10, 0.05), (30, 0.05), (30, 0.01)\}$ and across two detector architectures, their NRMSE differs from naive ranking by less than 0.001. With approximately 50 Gaussian fine-tuning samples, LISTA closes the gap, indicating that the zero-shot advantage is most relevant in the small-target-data regime.

TABLE I
ZERO-SHOT FOURIER $\to$ GAUSSIAN TRANSFER AT $T = 30$, $\lambda = 0.05$, $n = 256$, $m = 128$, AND $k = 25$.

Method	Gaussian NRMSE
Raw ISTA	0.4465
Naive top-k + LS	**0.2574**
Learned detector + LS	0.2578
LISTA, zero-shot	0.3267
LISTA, 50 Gaussian fine-tuning samples	0.2551
Oracle support + LS	≈ 0

III. Diagnosis: Support, Not Amplitude

The oracle-support result isolates the source of error. Once the correct support is known, support-restricted least squares gives nearly exact recovery. Therefore, the gap between the practical methods and oracle recovery is mainly caused by support misidentification rather than by inaccurate coefficient estimation.

The plateau of learned detectors suggests that the feature interface is too restrictive. The detector input at coordinate j is derived from $x_j^{(T)}$ and related magnitude statistics. However, $x^{(T)}$ is itself a nonlinear product of the ISTA dynamics and the sensing operator. Under a Fourier $\to$ Gaussian shift, these iterate-only features preserve the same coordinate-ranking information already exploited by naive top-k selection, but they do not explicitly encode the geometry of the deployment operator. This explains why the learned detector can match the magnitude baseline yet fails to improve over it.

IV. Toward Operator-Aware Support Detection

The diagnosis suggests that support detectors should not be restricted to iterate-derived magnitude features. For each coordinate j, we propose augmenting the detector input with operator-aware features computed at test time:

$$\phi_j = \left[|x_j^{(T)}|,\ A_{:,j}^\top (y - Ax^{(T)}),\ \|A_{:,j}\|_2,\ (A^\top A)_{jj},\ \max_{i\neq j} |A_{:,i}^\top A_{:,j}| \right]. \tag{3}$$

The residual correlation term measures whether coordinate j can still explain the current measurement residual, while the column norm, Gram diagonal, and coherence terms describe the local identifiability of that coordinate under the deployment operator. A lightweight coordinate-wise classifier can then estimate support probabilities from ϕ_j, after which amplitudes are computed by least squares on the selected support.

This design reintroduces the operator-dependent information exploited by unrolled methods such as LISTA, but confines learning to the support decision stage. The key empirical question is whether this operator-aware interface can improve zero-shot cross-operator support recovery and approach the performance of target-operator fine-tuning without using target labels.

V. Conclusion

Our results identify a support bottleneck in zero-shot cross-operator sparse recovery. In the studied Fourier $\to$ Gaussian setting, a simple ISTA + top-k + least-squares pipeline outperforms zero-shot LISTA, while learned iterate-only support detectors fail to exceed naive magnitude ranking. These findings motivate operator-aware support detection as a minimal learned module for improving transfer: rather than learning the entire inverse map, the model learns only the support decision using both iterate state and test-time operator geometry.